

Volunteer In-Kind Services — Technology Development

Amber's Angels, Inc. · EIN 42-2052151 · 103 Springwood Dr, Carrollton, GA 30117

CONTRIBUTOR Grant Lindberg	ROLE Founder / Lead Engineer
PERIOD OF SERVICE March 22 – May 15, 2026	HOURLY RATE (FAIR MARKET VALUE) \$150.00 / hr

DATE(S)	DESCRIPTION OF WORK	HOURS	VALUE
PHASE 1 — CORE MVP			
Mar 22	Initial system build RTMP ingest server, frame recording pipeline, OpenALPR integration — first working plate-detection loop.	4.0	\$600
Mar 23–25	MVP pipeline stabilization Package configuration, Python environment, process management, integration testing across ingest and detection layers.	11.0	\$1,650
Mar 26–27	Fixes and environment hardening Minor pipeline corrections, dependency resolution.	2.0	\$300
Mar 30–31	Monorepo + mission map + volunteer portal Consolidated frontend into monorepo under web/. Built mission map (Mapbox), pilot registration portal, Flock camera scraper, AggregationService for detection windowing, telemetry ingest endpoints.	9.0	\$1,350
Phase 1 Subtotal		26.0	\$3,900
PHASE 2 — DETECTION PIPELINE, AUTH & MOBILE APP			
Apr 1	FEMA IPAWS integration + full Expo mobile app Live FEMA CAP XML polling for AMBER/Silver/Blue/Mattie's/Purple alerts. YOLOv8 hybrid vehicle identification. Full Expo React Native mobile app: camera screen, GPS telemetry, detection	10.0	\$1,500

DATE(S)	DESCRIPTION OF WORK	HOURS	VALUE
	feed, mission map, leaderboard, mission debrief. FEMA polygon distance warnings. Auth/login wall.		
Apr 2	Auth, rate limiting, PM2, mobile auth Pilot registration approval flow, rate limiting, approval email, profile page, alert history, privacy/ToS pages. PM2 ecosystem config. Mobile auth + mission management UI.	5.0	\$750
Apr 3	DJI MSDK V5, push notifications, background scanning, badge system, E2E tests DJI MSDK V5 Kotlin native module (Avata/Mini3/Air3/Mavic3 camera + GPS). Android Foreground Service background phone scanning. Shared frame queue (one worker, all drones). Expo push token registration, watch-area push alerts, notification preference settings. FEMA polygon on map, tap-to-navigate. Pilot badge system (16 badges, 5 tiers). Self-service password reset via 6-digit email code. UI/UX polish pass. 31-test E2E suite for FEMA alert pipeline.	12.0	\$1,800
Apr 4	Frame preprocessing + composite confidence scoring Two-pass CLAHE + perspective deskew preprocessing pipeline. Composite confidence scoring with YOLO vehicle corroboration and vehicle profile matching. Discord image attachment on high-confidence hits.	5.0	\$750
Apr 5	Android compatibility Lowered minSdkVersion to 24 for broader Android device support.	1.0	\$150
Apr 6	DJI wiring, EAS OTA, coordinator role, adaptive frame rate Full DJI MSDK V5 Kotlin compilation and wiring. EAS OTA update integration. Adaptive phone frame rate driven by server detection signal. Mission debrief auto-selects from URL param. Coordinator role and flight priority zones. iOS build config. GPS coordinates in Discord alert embeds.	8.0	\$1,200
Phase 2 Subtotal		41.0	\$6,150
PHASE 3 — HARDENING, GRANT DOCUMENTATION & SSO			

DATE(S)	DESCRIPTION OF WORK	HOURS	VALUE
Apr 21	Alpha hardening + CDC classifier + grants directory Secrets moved to .env, structured logging, HTTPS configuration. CDC generational vehicle classifier cascade integrated. Grants directory created: MISSION_AND_IMPACT.md, initial grant strategy documents.	5.0	\$750
Apr 22	GitHub Pages, domain migration, HTTPS, grant documents GitHub Pages volunteer/privacy site. Migrated to amberangels.org. HTTPS enable script. Grant documentation: strategy, technical whitepaper, annual budget, SCORE financial template with SDVOSB in-kind estimates.	6.0	\$900
Apr 23	Public landing page, nginx, pitch deck Public marketing landing page at root route. nginx server block for Next.js. Pitch deck hosted at /deck. Navigation, email, and image path corrections.	5.0	\$750
Apr 24	Google + Apple SSO Complete SSO backend: Google OAuth, Apple Sign In, JWKS caching, cross-platform client IDs. Frontend nav wired to SSO state.	4.0	\$600
Phase 3 Subtotal		20.0	\$3,000
PHASE 4 — CI/CD, DEPLOYMENT INFRASTRUCTURE & COVERAGE ANALYSIS			
May 4	Pitch deck v1 + SSO security hardening Carrollton PD pitch deck. SSO hardening: eliminated f-string SQL injection risk, Apple JWKS caching, pending-account bypass fix.	4.0	\$600
May 8	Auth middleware refactor + mobile build Dropped legacy auth middleware layer, switched mobile to app-bundle output type.	1.0	\$150

DATE(S)	DESCRIPTION OF WORK	HOURS	VALUE
May 10–11	GitHub Actions CI/CD pipeline + infrastructure fixes GitHub Actions auto-deploy on push to main. nginx config corrections (www redirect, /api and /frames locations). SSH key/fingerprint setup and debugging. SVG explainer diagrams redrawn (DJI connection, camera angles). Landing page copy corrections.	8.0	\$1,200
May 12	Coverage grid, change password, RTMP settings, detection fixes Pilot dark-zone coverage grid with toggle controls. Change-password endpoint and settings UI. RTMP stream setup on mobile settings screen. Golden frame save path fix for Discord photo attachments. Alert dispatch wiring and VMMC confidence penalty.	6.0	\$900
Phase 4 Subtotal		19.0	\$2,850
PHASE 5 — FULL FEATURE SPRINT (APP STORE, SWARM, COVERAGE, DECKS)			
May 13	Flock Deadspace Planner, NCMEC poller, RBAC, heatmap Flock camera coverage planner with zip-code search and dark-zone visualization. NCMEC per-state RSS poller for case resolution detection. CAP Cancel handling (alert ended → watchlist deactivation). Role-based access control for coordinator alert resolution. Coverage heatmap on mission map. Google SSO integrated into map page.	5.5	\$825
May 14	App Store resubmission, account deletion, autonomous drone waypoint missions, FAA tiers App Store guideline compliance: camera pre-prompt text (5.1.1iv), account deletion flow (5.1.1v), law-enforcement framing corrections (2.1). Autonomous drone waypoint mission system: plan/dispatch/status lifecycle, VLOS/BVLOS/BVLOS-autonomous FAA operational tiers, mission timeout cleanup, observation post dispatch. Swarm drone heartbeat endpoint and mobile beacon. PM2 worker process management. can_dispatch_drones admin permission flag.	7.0	\$1,050
May 15	Swarm dispatch map UI, ADS-B airspace, road coverage, coordinator tools, volunteer	9.0	\$1,350

DATE(S)	DESCRIPTION OF WORK	HOURS	VALUE
	stories deck Swarm drone dispatch UI on mission map (drone icons, dispatch modal, observation point auto-fill). ADS-B IN live aircraft layer with airspace deconfliction warnings. Road segment heatmap replacing grid squares; coverage auto-seeded on every new FEMA alert. CI self-healing (root-owned node_modules fix, passwordless sudo). Per-coordinator SMS opt-in. Mission map sidebar overhaul with NCMEC case feed tab. Coordinator dispatch mobile screen with role-aware tab navigation. Admin pending users section. Volunteer stories deck (4-persona: phone-mounted, Part 107 VLOS, gig driver, BVLOS swarm). Test alert push notification pipeline. Worker idle CPU fix. Privacy language across stories deck.		
Phase 5 Subtotal		21.5	\$3,225
TOTAL IN-KIND CONTRIBUTION		127.5	\$19,125

Basis for fair market value: Rate of \$150/hr reflects fair market value for senior independent contractor services encompassing full-stack web development (Next.js, FastAPI, PostgreSQL), cross-platform mobile development (Expo React Native, Android Kotlin native modules), AI/ML pipeline integration (YOLOv8, OpenALPR, Plate Recognizer), cloud infrastructure and DevOps (DigitalOcean, PM2, GitHub Actions, nginx), and systems architecture. This rate is consistent with independent contractor market rates for equivalent expertise in the Atlanta, GA metropolitan area and nationally.

In-kind contribution documentation: Work is documented in full via the project git repository (github.com/grantl12/ambers-angels). Each line item corresponds to one or more dated commits. This timesheet may be used for IRS Form 990 Schedule O volunteer service documentation, grant application in-kind match calculations, and internal organizational records. Note: volunteer services are not deductible as a charitable contribution under IRC § 170, but the fair market value of donated services is recognized as an in-kind contribution for grant matching purposes per 2 CFR § 200.306.

Organization: Amber's Angels, Inc. — 501(c)(3) applied, EIN 42-2052151, Carroll County, GA pilot program.

Grant Lindberg

Founder / Lead Engineer, Amber's Angels

Signature

Date

Matthew J. Stone

Director, Amber's Angels, Inc.

Signature

Date

Justin P. Espinosa

Director, Amber's Angels, Inc.

Signature

Date